Problem Statement

Mobile App User Engagement and Subscription Analysis

Project Context

Mobile applications generate interaction data across users, devices, sessions, event types, and locations. Understanding these interactions can help a product team describe how users engage with the app and how engagement patterns differ by operating system, device model, country, and subscription status.

Problem

The project analyzes a dataset of mobile app user interactions to understand common engagement behaviors and explore how interaction activity relates to subscription status and device or network characteristics. Before analysis, the data must be inspected and cleaned so that missing values, duplicate records, inconsistent data types, and outliers do not unnecessarily undermine the results.

Project Objectives

- Inspect and clean the mobile app interaction dataset, including handling missing values, data types, and outliers.
- Summarize interaction patterns across event types, event targets, operating systems, device models, and countries.
- Describe the distribution of subscribed and non-subscribed users and compare subscription rates across available categories.
- Explore numerical fields such as battery level, memory usage, event value, session duration, and user age.
- Present the findings in a Power BI dashboard to support clear, accessible exploration of the data.

Analytical Scope

The source documentation reports 100,000 raw records and 21 columns. After cleaning, the dataset contained 98,656 records and 21 columns. The analysis is descriptive and identifies patterns in the available data; it does not establish that any device, event, or user characteristic causes a change in subscription behavior.

Expected Outcome

A cleaned dataset, exploratory analysis, and a two-page Power BI dashboard that summarize user interactions and subscription-related patterns, while highlighting areas for further investigation.